PREET PATEL

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Experience

U.S. Patent & Trademark Office · *Patent Examiner, Photonics & Optical Systems* *Oct 2024 – Present*

- *Technical Architecture & Prior-Art Analysis.* Evaluate complex optical circuitry, semiconductor, and photonics systems against prior-art literature and statutory compliance standards; authored **100+ technical office actions** under strict production quotas.

Mercor · *AI Evaluation & Model Quality Specialist (Contractor)* *Apr 2026 – Present*

- *Frontier Model Evaluation.* Execute high-volume human-in-the-loop evaluations and rubric-based audits of frontier LLM outputs for multi-step reasoning, factual grounding, and instruction adherence; selected for priority fast-turnaround batches based on high inter-annotator agreement and diagnostic precision in surfacing model failure modes.

GlobalLogic Inc. (Google RMI Team) · *LLM Quality & Safety Analyst* *Jul 2024 – Oct 2024*

- *Adversarial & Edge-Case Auditing.* Conducted structured evaluations of Gemini and Magi search/reasoning outputs against safety and truthfulness guidelines, categorizing hallucination patterns, logic breakdowns, and data anomalies.

Education

University of California, Davis · *M.S., Physics* *2020 – 2023*
University of Michigan, Ann Arbor · *B.S., Physics; B.S., Astronomy & Astrophysics; Minor in Statistics* *2015 – 2019*

Research Experience

Type Ia Supernova Delay-Time Distribution Inference · *Independent Research (FIRE Collaboration)* *2026*

- Built a Bayesian inference pipeline comparing ten delay-time-distribution families (power-law through skewed-peak models) against Milky Way stellar abundance data from APOGEE DR17, using the element-tracer method on FIRE-2 cosmological simulations.
- Wrote a factorized yield integrator approximately 25× faster than the original numerical integration while agreeing to about one part in a million.
- Maintains a fork of the FIRE analysis code (GizmoElementTracers) with MCMC-based parameter estimation over stellar [Mg/Fe]–[Fe/H] distributions.

Elemental Abundances of Simulated Low-Mass Galaxies · *UC Davis, Computational Astrophysics* *Mar 2021 – Sep 2023*

- Engineered reproducible Python and HDF5 data pipelines for >**1 PB** FIRE-2 cosmological simulation datasets on national supercomputing clusters, cutting runtime by **50%+** via vectorization, redundancy removal, and parallelization.
- Identified elemental abundance signatures of bursty star formation and satellite accretion in present-day abundance distributions of low-mass galaxies.
- Built likelihood-based parameter estimation workflows (MCMC code) and gradient-descent models on noisy, high-dimensional (20+ feature) datasets to isolate statistically significant signals and quantify confidence bounds.
- Led end-to-end research from pipeline design to first-author publication in *MNRAS*, 512(4), 5671, collaborating across a 300+ scientist consortium with shared version-controlled software.

Projects

Rust Data Workbench · *Local-First CSV Exploration & Classical ML* *Oct 2026*

- Scoped a native Rust (egui, Polars, SmartCore candidates) alternative to notebook-stack data work: CSV import, data profiling, reversible cleaning/subsetting, interactive visualization, and supervised classification/regression with reproducible splits; local-only, no telemetry, source CSV never overwritten.
- Test-first build under a supervised multi-model pipeline: an 8-phase plan with a separate author and reviewer per slice, and a data contract plus an 85-file red-test bundle (27 data-import behavior tests written to fail against stubs) refined through nine independently reviewed revisions. No app code applied yet; the latest patch gate is held on build-script sandboxing fixes before implementation begins.
- *Planned.* CSV/profile core, transform engine, desktop shell, charts, ML pipeline, then end-to-end smoke tests and independent UX/code review.

Agent Workbench · *Tauri · Rust · Svelte · MCP · mouse-friendly tmux for AI-agent supervision* *Oct 2026 – Present*

- Built a Windows desktop workspace on a persistent terminal service (broker daemon plus one host process per terminal): terminals survive closing the app, restarting the broker, and rebuilds, and are re-adopted on restart. Implements tmux's default prefix keys (split tree to 64 panes, repeat-resize, zoom, copy mode, scrollback search, synchronized input, output logging), plus a Canvas of floating terminals and a scriptable `wb` CLI.

- Observer and Inspector read Claude Code, Pi, and Codex transcripts to show every agent session's state, tools, usage, and model, with an agent map; agents may only propose commands, and type into a terminal under time-limited human grants (dormant when no human is attached).
- Hardened the service (current-user-only DACL, HMAC challenge with DPAPI-protected secret, elevated clients refused, strict clippy gate, Pi/Argon adversarial review with all findings fixed); an 8-hour, 46k-round soak test traced memory growth to a per-connection security-descriptor leak, fixed and re-verified; an end-to-end test proves detach, broker restart, re-attach, grants, and full process teardown. Role-split governance: Claude designs/reviews, DeepSeek makes volume edits, guard.py gates destructive actions (allow/deny/escalate), with per-task evidence in a versioned handoff tree.
- *MCP Server & Shared Agent Board.* Built `wb-mcp`, a Rust MCP server (6 tools) registered in Claude Code, Codex, Pi, and Antigravity, so agents from different vendors can see each other's sessions and coordinate. Tools expose read-only session metadata plus a shared board (presence, file claims, notes, done, and ask posts) stored by the daemon in SQLite and served over a separate named pipe whose access rules admit sandboxed agents without widening the main service.
- *Injection-Safe Coordination.* Board posts reach agents as framed, escaped data, never instructions, with invisible and format Unicode characters stripped and every field bounded; clients verify the identity of the process serving the pipe before connecting. Lifecycle hooks post presence at session start and end, refuse an edit once when another session holds a live claim on that file (PreToolUse), and post a `done` note on each git commit. Each slice passed independent multi-round reviews (all findings fixed) before merge.
- *Planned.* Real sign-in and credential handling for other users, packaging and signing, and a free, bring-your-own-key release.

MCP Servers & Claude Code Extensions · *Python · TypeScript · Model Context Protocol* *Sep – Oct 2026*

- Wrote a Python MCP server that exposes Gemini as a tool to Claude agents, with per-call model selection and a usage and cost log.
- Integrated a local PyTorch text classifier (Laya, Hugging Face checkpoints) into Claude Code as an MCP tool server; moved it to CUDA (∼**9**× faster per question) and tuned startup so health checks fit the client's 30 s timeout.
- Built a Claude Code plugin on lifecycle hooks (39 tests) with a Deck pane showing files touched, tool calls, per-task cost, and a model switcher. Maintain shared agent skills and global rules synced one way from a single source to five agent harnesses.

Prompt-Injection Red Team · *LLM Security Evaluation · Python* *Oct 2026*

- Before letting an unattended agent read untrusted job-posting text, tested an extraction prompt against 11 injection payload classes plus 2 benign controls, across three tool-access configurations (25 runs, harmless canaries). Result: **0/25** compliance and 25/25 schema-valid output.
- Caught a false positive in the automated judge (it matched on text, not behavior) by reading raw outputs, and concluded that safety must be structural (no tools, schema-validated output) rather than left to the model.

Agent Dashboard · *Live Visualization of AI Agent Activity · Python · JavaScript* *Oct 2026*

- Built a local web view that turns Claude Code transcripts into a live map of agents, the tools and documents they touch, and who reports to whom, with a JSON roster contract any orchestrator can write to name agents and draw org-chart edges. Includes a simulated newsroom ($0.09–0.13 per live run) and a mock mode that generates synthetic transcripts for 100+ agents.

patelpb.com · *Personal Site & Interactive Data Visualizations · React · AWS · live* *2026*

- *ATP Top 20 Explorer.* Interactive chart of decades of men's tennis rankings: a Polars pipeline computes top-10 metrics and appends new weekly lists scraped with headless Chrome; features include player groups and presets, moving averages, a career-comparison mode aligned by debut date, drag-to-reorder layers, and 293 processed player portraits (background removal and face crop).
- *Market Explorer.* A browser-side analytics page over market price history stored as 256 Parquet shards and read client-side, with derived metrics, threshold filters, a search syntax, and a scheduled Lambda refresh.
- *Infrastructure.* Migrated hosting to S3 and CloudFront with Route 53 DNS and CloudFront Functions for routing and edge authentication; serverless APIs on Lambda (AWS CDK) with cooldowns, locks, and a budget cap; a password-gated web search page backed by the Exa API. Research page shows ten animated model fits from the supernova work.

R2S2 · *Godot 4 .NET (C#) Skating/Movement Game* *Oct 2026*

- Rebuilt character movement as a JSRF-style surface state machine: a probe-validated surface frame (seven point probes, facet-spike-filtered normals) replaces rigid-body ground adhesion; tangent-plane RideMotion holds a 49° pipe shell with no minimum speed. Probe-driven movement is now the default, with the legacy controller kept selectable for comparison.
- Decoupled animation from physics with an analytic two-bone leg solve; deterministic `verify.sh` gate runs the 59-check friction suite (30/60/120 Hz), BotQC (50 checks), and the 83-check legacy suite green before any merge.
- Root-caused and fixed a lateral-friction bug (the scrub projected onto an axis built from the velocity it measured, so it was identically zero) and a steep-pipe sideways-landing bug (stale takeoff normal rejected the floor); added a route-clearance gate and rig-clearance measurement so fixes are tied to actual mesh clearance, not origin position.

- *Planned.* Human playtest of landing and carving feel, rails as curve traversal and wall ride as plane traversal at parity with legacy, then flip default and retire legacy; design specs written for graffiti/notoriety, equipment and skills, death/loss, and MMO networking and hosting.

Agent Report · *Multi-Agent LLM Newsroom* *Oct 2026*

- *Multi-Agent Architecture.* Designed and shipped a hierarchical multi-agent system that researches, writes, fact-checks, and publishes a daily news edition: a frontier-model (Claude Opus) editor-in-chief delegates to Sonnet senior editors and Haiku reporters through assign, review, revise, and publish loops run headlessly by a Python daemon.
- *Cost & Runtime Optimization.* Cut per-edition cost **46%** ($1.08 → $0.58) and paid web searches **53%** (55 → 26) via a shared research desk that caches/deduplicates sources, prioritizes free RSS feeds before paid search, and tiers model routing; full editions compile in 2–3 minutes.
- *Offline Eval Harness & Local Classifier* (*Laya*). Built an offline evaluation harness for a local relevance classifier screening search candidates before LLM ingestion (571 labeled candidates across 8 editions; 4 prompt templates compared via ROC-AUC and precision/recall cutoffs). The winning headline-only template (**AUC 0.96**) prunes **59%** of noise at 100% production recall.
- *GPU Inference Acceleration.* Migrated local classifier inference from CPU to CUDA (RTX 4060 Ti), cutting scoring latency ∼**17**× (406 → 23 ms/candidate); gated rollout behind shadow-mode logging to monitor drift.
- *Cloud Infrastructure & Guardrails.* Enforced a declarative source policy (≥2 independent sources for major claims, automated confidence labels for single-source signals). Deployed delivery over an encrypted AWS relay (**Lambda, CloudFront edge auth, S3, SSM**).

Publications

- S. Chughtai, S. Kamath, K. Patel, **P. Patel**, M.U. Ghani. "Predictors and contributors of burnout among hospitalists in central Michigan." *Journal of Healthcare Risk Management* (2026). PMID: 42307986
- **P. Patel**, S. Loebman, A. Wetzel, C.-A. Faucher-Giguère, K. El-Badry, J. Bailin. "Predictions for Complex Distributions of Stellar Elemental Abundances in Low-Mass Galaxies." *Monthly Notices of the Royal Astronomical Society*, 512(4), 5671 (2022). doi:10.1093/mnras/stac834

Skills

AI & LLM Systems	Multi-Agent Orchestration, Model Context Protocol (MCP), Agent Hooks & Plugins, Anthropic API (Opus/Sonnet/Haiku), Gemini API, Offline Eval Harnesses, Prompt Engineering & Ablations, Prompt-Injection Testing, PyTorch, Hugging Face, CUDA, Rubric & Guardrail Design
Cloud, Data & Infra	Python, R, SQL, SQLite, Rust, C++, C#/Godot, JavaScript/React, Tauri/Svelte, AWS (Lambda, S3, CloudFront, Route 53, SSM, CDK), Polars, Pandas, NumPy, SciPy, HDF5, Parquet, Linux/Unix, Git
Visualization	Data Visualization, Interactive Web Charts, Matplotlib, Microsoft PowerPoint, Excel
ML & Statistics	ROC-AUC / Precision-Recall Optimization, Classification, Statistical Inference, MCMC, Maximum Likelihood, Nonparametric Tests, Regression, Time-Series Analysis, Uncertainty Quantification